Asset Engineering – Week 6

Group	Group 2 - AI Exam Answer Grader
Roll Numbers	J008, J009, J010, J011, J012
Class & Course	B.Tech DS 4th year – Speech & Natural Language Processing (SNLP)

1. Objective

The Week 6 target was to convert the NLP model built in Weeks 1–5 into a reusable component: an API/module, structured input/output, documentation, and confidence scores where applicable.

Before building that asset, the following prerequisite work was completed, without modifying the EngSAF dataset or any previously submitted file:

- Reviewed Weeks 1–5 code for defects and certain limitations.
- Improved model accuracy, validated honestly (every choice made on the validation set, test set touched once).
- Re-ran the Week 5 evaluation protocol on the improved model.

Then, after making the necessary changes, we built our reusable asset (API Service).

2. Bug Fixed (Week 1-5)

- Score scaling used Python's built-in round(), which applies banker's rounding (round-half-to-even), e.g. round(2.5) returns 2, not 3. This silently under-scores answers whenever (label/2.0) × maximum_marks lands exactly on a half-point. Fixed with math.floor(x + 0.5), which always rounds half up.

3. Accuracy Improvement (V3)

The Week 5 error analysis separately identified two concrete, addressable failure patterns:

- "Partial → Correct" was the single largest error type (93 of 240 test errors): an answer missing some reference content was still graded fully correct, because keypoint_coverage averages per-point similarity, letting one missing point get diluted by the other, well-covered points.
- Negation/contradiction was missed entirely, e.g. a student answer of "True" against a reference answer of "FALSE" still scored as a correct match on shared vocabulary alone.

Changes:

V3 keeps every Week 4 (V2) feature except keypoint_coverage, which is replaced by keypoint_min_coverage (the worst-covered reference key point, instead of the mean), and adds a new negation_mismatch feature (flags when exactly one of the two answers contains a negation cue word).

Model Selection:

Following the same honest protocol as Week 4: every candidate model is scored on the validation set; the test set is touched exactly once, after the best candidate is chosen. Logistic Regression (various C / class_weight), Random Forest, and Gradient Boosting were tried; a tuned Logistic Regression (C = 1.0, class_weight = None) was again selected.

Model	Accuracy	Macro-F1	QWK	MAE	Partial recall
V2 — Hybrid features + LR (Week 4)	0.5257	0.5005	0.3643	0.5711	0.3770
V3 — keypoint_min_coverage + negation (this pass)	0.5494	0.5252	0.3919	0.5415	0.4084

4. Week 5 Re-Run on V3

The Week 5 evaluation protocol (confusion matrix, error-type breakdown, confidence analysis) was re-run against V3 on the same, untouched test set.

Rows = true label, columns = predicted label

	Pred: Incorrect	Pred: Partial	Pred: Correct
True: Incorrect	45	31	37
True: Partial	23	78	90
True: Correct	9	38	155

Total errors fell from 240/506 (V2) to 228/506 (V3). "Partial → Correct", the error type the V3 features directly targeted, fell from 93 to 90 cases — a modest but real reduction, consistent with the metric gains above. Confidence remains informative but not sufficient on its own to flag every wrong prediction: mean confidence was 0.582 for correct predictions vs. 0.516 for incorrect ones, with overlapping ranges.

5. Asset Architecture

```
Input (Question, Reference Answer, Marking Rubric, Student Answer, Max Marks)
        |
        v
Preprocessing  (clean_text: lowercase, strip punctuation)
        |
        v
NLP Model      (V3: MiniLM semantic similarity + TF-IDF + key-point
                min-coverage + word overlap + length features +
                negation mismatch  ->  tuned Logistic Regression)
        |
        v
Post-processing (score scaling, missing-key-point feedback text,
                 confidence = max class probability)
        |
        v
API / Asset    (FastAPI: POST /evaluate-answer)
        |
        v
Application    (e.g. faculty grading dashboard; consumes Group 1's
                OCR output directly as student_answer)
```

6. API/ Module Documentation

a) Endpoint

- **POST /evaluate-answer**

Query parameter model (optional, default v3): one of v3, v2, v1, baseline selects which trained approach grades the answer.

- **GET /health**

This reports load status and available model versions.

b) Input

```
{
  "question": "Q4 What does supervised learning use to train model?",
  "reference_answer": "A supervised machine learning algorithm relies on
                       labeled training data to learn a function that maps
                       inputs to outputs.",
  "marking_rubric": null,
  "student_answer": "Supervised learning uses labeled data to train models
                     to predict outcomes.",
  "maximum_marks": 10
}
```

c) Output


```
{
  "question": "Q4 What does supervised learning use to train model?",
  "score": 10,
  "maximum_score": 10,
  "feedback": "The answer covers all core concepts present in the reference.",
  "confidence": 0.62,
  "missing_key_points": [],
  "model": "v3_improved"
}
```


d) Example invocation : partial-answer case

V2 originally graded it 6/6 which is fully correct, missing both omitted points entirely. V3 correctly identifies it as partial.

```
curl -X POST "http://localhost:8000/evaluate-answer" \
  -H "Content-Type: application/json" \
  -d '{
        "question": "Q2 Privileged operations",
        "reference_answer": "- Write to CR3 register to modify page directory
                              pointer\n- Write to the IDTR register to change
                              location of IDT\n- Access to VA that lies in the
                              kernel virtual address range",
        "student_answer": "Writing to the CR3 register to change the page
                           directory.",
        "maximum_marks": 6
      }'
```


```
{
  "question": "Q2 Privileged operations",
  "score": 3,
  "maximum_score": 6,
  "feedback": "The answer is partially correct but misses some key concepts.
              Check these points: write to the idtr register to change
              location of idt; access to va that lies in the kernel virtual
              address range.",
  "confidence": 0.46,
  "missing_key_points": [
    "write to the idtr register to change location of idt",
    "access to va that lies in the kernel virtual address range"
  ],
  "model": "v3_improved"
}
```


7. Reusability – Incorporating Group 1's OCR output

Group 1 (OCR / Handwriting Extraction) outputs extracted text plus a confidence score for low-confidence regions. This asset's student_answer field accepts that output directly: pass Group 1's extracted text as student_answer, unchanged. No adapter is needed, because both assets already speak plain text with a confidence field in their output — the intended interoperability model for the AI University Platform.

8. Limitations

- negation_mismatch is a coarse word-list heuristic, not true negation-scope detection that means it can miss negation expressed without a listed cue word, or false-positive on an unrelated use of one.
- Small test set (506 examples): V3 is a real, validation-selected improvement over V2, but all four models sit in the low-to-mid 50% accuracy range on a genuinely hard three-class ordinal problem, not a solved one.

9. Conclusion

A targeted accuracy-improvement pass (V3), directly motivated by the Week 5 error analysis, improved every reported metric over the Week 4 model (accuracy 0.5257 → 0.5494; Macro-F1 0.5005 → 0.5252), validated honestly against the test set exactly once. A further experiment with a stronger embedding model was tried, found inconclusive, and deliberately not adopted but documented rather than discarded. The resulting V3 model is now wrapped as a reusable FastAPI asset (POST /evaluate-answer) ready to be consumed by another application in the AI University Platform, such as Group 1's OCR output feeding directly into student_answer.